EXECUTIONPROOF_PIN — EP-AGENT-001

EXECUTIONPROOF PIN — EP-AGENT-001 (checklist item 7)

```
DESIGN:                         COMPLETE
EXECUTIONPROOF PIN:             PASS
Q9:                             PASSED
Q8 TECHNICAL FREEZE READINESS: PASS
Q8 FORMAL:                      OPEN — REQUIRES DEREK APPROVAL + PREREGISTRATION FILING
PREREGISTRATION:                NOT FILED
EXPERIMENT:                     NOT EXECUTED
RESULTS:                        NONE
EXECUTION:                      NOT AUTHORIZED
```

Current pin — v0.1.2-reference (Derek Hone decision: option (a), 2026-10-16)

Status: PINNED — CHECKLIST ITEM 7 PASS (technical pin). This section supersedes the `v0.1.1-reference` candidate pin for EP-AGENT-001; earlier sections are retained unedited, in chronological order, under HISTORICAL / SUPERSEDED below. Derek Hone chose option (a): a separate, hashed, fail-closed evidence registry file. Study policy SHA-256 is unchanged.

Field	Value
Implementation name	ExecutionProof Controlled Reference Verifier
Repository (local, no remote)	`/home/ubuntu/executionproof-reference-verifier`
Branch	`main`
Tag	`v0.1.2-reference` (annotated). Earlier tags retained: `v0.1.1-reference` → `279bcb2669e4602ae25e0d745d827a1c8514265e`, `v0.1.0-reference` → `a578823eb686424d5fa39786fbe594d297a2dde6`
Commit	`b03365d57880e896c35e72898f8335a1826afe74`
API / implementation version	`0.1.2-reference`
Verify	`POST /verify` → as v0.1.1 plus `evidence_registry_hash`
Proof	`GET /proof/{id}`
Health	`GET /health` → `status`, `implementation_version`, `git_commit`, `policy_hash`, `evidence_registry_hash`, `evidence_registry_id`, `evidence_registry_version`, `proofrecord_schema_version`
15-field EP-AGENT-001 contract accepted directly	YES (unchanged)
Policy loading	`EP_POLICY_CONFIG` (absolute path), fail-closed (exit 1). A policy containing `rules.evidence_records` is now **rejected** (fail-closed)
Evidence loading	`EP_EVIDENCE_REGISTRY` (absolute path), fail-closed (exit 3) if missing, unreadable, relative, malformed, wrong schema, duplicate reference/id, or invalid status
Evidence resolution rule	A reference counts only on an exact string match to a registry record whose `status` is exactly `VERIFIED`. Unknown reference → HOLD with `EVIDENCE_REFERENCE_UNKNOWN`. Reference strings are never parsed or inferred from

Field	Value
Study policy path / SHA-256	`/home/ubuntu/EP-AGENT-001/fixtures/policy/fixture_policy_config.json` — `06b5a7ab68696d4d246f7b07f698437ec5713fa1da1113f553a02e04777649d4` (unchanged)
Study evidence registry path / SHA-256	`/home/ubuntu/EP-AGENT-001/fixtures/evidence/study_evidence_registry.json` — `b4a84f293182cd299b9168f3d38f3fc8f00e56c6bc89bcfa74a6385a490747d0`
Study registry identity	schema `ep-evidence-registry/1`, id `EP-AGENT-001-STUDY-EVIDENCE-REGISTRY`, version 1, 2 records (`EP-AGENT-001-EVR-001`, `EP-AGENT-001-EVR-002`, both `VERIFIED`)
ProofRecord schema	version 2 (19 columns; adds evidence registry hash, id, version, and schema version). Store refuses schema-1 databases; a fresh DB file is required
Implementation config hashes	`requirements.txt` `583da51cb298743be1d697178aa38a2b00fbd1473dd28f1fb5444e266d050eb3`; `pytest.ini` `8fa307d1ef787f3822b949dc90f1b3c405769c4eb97bb2e813b271109d9d8a90`; `.gitignore` `d6235bec62a32ab242c2f52a835bc50f2da3ff5202b4a6f06c88e9a4241b7184` (all unchanged)
Development-only artifacts (not used for the study)	`tests/fixtures/dev_policy.json` `d7a6256d0ed14d1b31bfc92847d70eca4449a8eee1e36b1679ca8c791bc9b0f4` (changed: `evidence_records` removed); `tests/fixtures/dev_evidence_registry.json` `bbf199b7853de91c69fbe367bd4d7bff9515083dd09317b783a9102ad0e2a048` (new)
Verifier `MANIFEST.sha256` SHA-256	`e53564589c4baeaca7bb1e26277c79abcb3f9e012b2f17fb26be19005e11eb69` (36 entries; `sha256sum -c` OK)
Dependency lock SHA-256	`requirements.lock` `1e4a988c867df880a23e1586d6d5e3f9340b447dd232a6ee2f9ba903a5de8702` (unchanged since v0.1.0-reference)
Development test suite	150 passed, 1 warning (DEV-REF inputs only; not EP-AGENT-001 evidence)
Load-only startup check	2026-10-16T02:13:12Z, clean tagged commit, `EP_POLICY_CONFIG` = study policy, `EP_EVIDENCE_REGISTRY` = study registry, throwaway DB; only `GET /health` called → `implementation_version` `0.1.2-reference`, `git_commit` `b03365d5…`, `policy_hash` `06b5a7ab…77649d4`, `evidence_registry_hash` `b4a84f29…490747d0`, `proofrecord_schema_version` 2; server stopped; 0 ProofRecords; 0 POST requests; throwaway DB deleted; no study fixture used
Runnable / Reference / Production	YES / YES / NO

How the study registry was built (static inspection only). The `contract.evidence_references` of B1–B5 were read from the JSON files: B1, B2, B5 → `evidence://verified/test-actor-authorized/identity-check-001`; B3 → `evidence://verified/test-actor-unauthorized/identity-check-001`; B4 → `[]` (intentionally no registry record). Each of those fixtures' `fixture_policy_context.evidence_records` lists the same reference with status `verified` (lowercase); the registry records it in the verifier's canonical status vocabulary as `VERIFIED`. The verifier does not read `fixture_policy_context`; at run time evidence status comes only from the hashed registry. No fixture was submitted to `/verify`; no expected verdict was used to choose registry content beyond these transcribed statuses.

Boundary. The registry provides deterministic benign evidence status only. It does not model freshness, revocation, provenance, replay resistance, tampering, or authenticity (out of scope; university-reserved lanes untouched).

Required disclosure (verbatim, unchanged):

> "The ExecutionProof implementation used in EP-AGENT-001 is a controlled reference implementation built and frozen before preregistration filing and before experimental execution. EP-AGENT-001 therefore evaluates cross-framework integration portability against this pinned reference implementation, not against an existing production deployment."

HISTORICAL / SUPERSEDED

Retained unedited in chronological order (Rule 11). Nothing in this section is current status. Superseded by the Current pin above.

Historical status (v0.3, superseded): BLOCKED — DEREK CONFIRMATION REQUIRED. No implementation commit has been invented.

What could be pinned (verified 2026-10-15)

Item	Value	Source
B1–B5 policy config	`fixtures/policy/fixture_policy_config.json`	extracted canonically (sorted JSON) from the shared `fixture_policy_context` of B1–B5; identical across all five
Policy config SHA-256	`06b5a7ab68696d4d246f7b07f698437ec5713fa1da1113f553a02e04777649d4`	`sha256sum`
Public contract reference (reference only)	`github.com/derekhone/executionproof`, branch `main`, HEAD `886b5c4b69ce4bc3793d51a075e0b986372ce61b` (2026-08-21); tag `v1.0.0-public` → `3aa3b15a955ebd23d9414fa9a111a8c5b550c502`	shallow clone
Public OpenAPI	`docs/openapi.yaml`, `info.version` 2.0.0 (OpenAPI 3.0.3), sha256 `1d65ca5a172df4367ab216ff02e0375ba71733062381dd444febd566fdd66651`	file in public repo

What could NOT be pinned — and why

The public repository contains specifications, docs and examples only — **no runnable verifier implementation**. The controlled implementation is private and not accessible from this environment. Discrepancies to resolve:

Packet assumption	Public OpenAPI 2.0.0
API family "v2.1"	`info.version: 2.0.0`
`POST {EP_BASE_URL}/verify`	`POST /v2/boundaries/{boundaryId}/verify`
`GET /proof/{id}`	`GET /v2/proofrecords/{proofId}`
15-field Request Contract body	`VerifyRequest {action, principal, evidence, requestedAt}`
Policy load via `/policies/*`	not present in public OpenAPI

Derek must supply (one line each)

1. ExecutionProof implementation repo + branch/tag + full commit SHA to run against.
2. API version and exact verify / proof-resolution paths (adapters accept `EP_BASE_URL` + `EP_VERIFY_PATH`; default `/verify`).
3. Whether that build accepts the frozen 15-field Request Contract as-is (if it requires a mapping layer, that layer is a pre-filing amendment, never an in-run change).
4. Configuration files used by that build, and how `fixture_policy_config.json` is loaded (their SHA-256s get recorded here).

v0.3.1 private-implementation hunt (2026-10-16) — STILL BLOCKED

Read-only search; no endpoint called with any B1–B5 fixture; nothing invented.

Source	Result
Local filesystem (`/home/ubuntu` incl. hidden/config dirs, cloned repos, `.aws`, `.config`)	No `/verify` or ProofRecord-retrieval server; no `policies/authority`, `policies/constraints`, `policies/evidence-verifiers`, `policies/rails`, `evidence-schemas`, `v2.1`, uvicorn/gunicorn service, or ExecutionProof Dockerfile/docker-compose; no `EP_BASE_URL`/ExecutionProof host in env
GitHub (Abacus GitHub App)	`GET /user/repos?visibility=private` → **0 repos**. App sees 18 public repos only. Private repos are **not visible**, which is not evidence that they do not exist
Hosting clues	`builderexecutionproof.io` (Abacus-hosted web app): HTTP 200 site, but `/api/verify`, `/api/v2/verify`, `/verify`, `/api/health`, `/api/proofrecords`, `/openapi.json` → 404; site states ProofRecords use "a lightweight deterministic fingerprint for local testing" → demo, not the controlled build. `executionproof.io/v2/proofrecords/x` → 404; `api.executionproof.io/health` → 404
Gmail (deploy/host/GitHub notifications)	No ExecutionProof API host, deployment, build ID, or private-repo notification found
Abacus projects	Only "Execution Boundary Weekly Production" AI_AGENT projects (content cadence), not a verifier

Excluded as backend (§6 of the hunt instruction): ARK-498 server, experiment-local gateways, EP-PQC harnesses, vaultproof-agent-guard, public OpenAPI spec alone.

Smallest Derek action: state where the runnable ExecutionProof v2.1 build lives: either (a) grant the Abacus GitHub App access to that private repo and give its owner/name, or (b) state that no runnable v2.1 build exists yet.

Derek Hone ground truth (2026-10-15)

Derek Hone's statement, verbatim:

> "I, Derek Hone, confirm that I do not know of an existing runnable private ExecutionProof v2.1 verifier implementation that can currently be identified and pinned for EP-AGENT-001."

Note: this statement records that no runnable private v2.1 verifier can currently be identified and pinned. It does **not** mean that no prior ExecutionProof implementation ever existed.

Candidate pin — ExecutionProof Controlled Reference Verifier for EP-AGENT-001

Status: BUILT — CHECKLIST ITEM 7 REMAINS OPEN pending Derek decision on evidence-reference status mechanism (options a/b/c)

Field	Value
Implementation name	ExecutionProof Controlled Reference Verifier
Repository (local, no remote)	`/home/ubuntu/executionproof-reference-verifier`
Branch	`main`
Tag	`v0.1.1-reference` (earlier tag `v0.1.0-reference` → `a578823eb686424d5fa39786fbe594d297a2dde6` retained)
Commit	`279bcb2669e4602ae25e0d745d827a1c8514265e`
API version	`0.1.1-reference`
Verify	`POST /verify` → `verdict`, `proof_id`, `proof_record_id` (= `proof_id`), `timestamp` (ISO 8601 UTC = ProofRecord `created_at`), `proof_url`, `execution_authorized`, `reason_codes`, `implementation_version`, `policy_hash`
Proof	`GET /proof/{id}`

Field	Value
Health	`GET /health` → `status`, `implementation_version`, `git_commit`, `policy_hash`
15-field EP-AGENT-001 contract accepted directly	YES
Policy loading mechanism	`EP_POLICY_CONFIG` env var (absolute path); fail-closed — process refuses to start (exit 1) if the file is missing, unreadable, relative, or invalid
Study policy expected path	`/home/ubuntu/EP-AGENT-001/fixtures/policy/fixture_policy_config.json` (not copied into the verifier repo)
Study policy SHA-256	`06b5a7ab68696d4d246f7b07f698437ec5713fa1da1113f553a02e04777649d4`
Implementation config hashes	`requirements.txt` `583da51cb298743be1d697178aa38a2b00fbd1473dd28f1fb5444e266d050eb3`; `pytest.ini` `8fa307d1ef787f3822b949dc90f1b3c405769c4eb97bb2e813b271109d9d8a90`; `.gitignore` `d6235bec62a32ab242c2f52a835bc50f2da3ff5202b4a6f06c88e9a4241b7184`; development-only policy `tests/fixtures/dev_policy.json` `423ba050de874785d5c4920a78b029784995e1f9705cc97b5a0390f57bab84c5` (not used for the study)
Verifier `MANIFEST.sha256` SHA-256	`09a78d7d31ce219f19cdc90c4180e8416c4dc98d302ed6b8694779bd6912323f` (32 entries)
Dependency lock SHA-256	`requirements.lock` `1e4a988c867df880a23e1586d6d5e3f9340b447dd232a6ee2f9ba903a5de8702` (unchanged from v0.1.0-reference; Python 3.11.6, fastapi 0.143.2, pydantic 2.14.1, uvicorn 0.54.0)
Development test suite	111 passed, 1 warning (DEV-REF inputs only; not EP-AGENT-001 evidence)
Load-only startup check	Server started at the tagged commit with `EP_POLICY_CONFIG` = study policy path; only `GET /health` called → `policy_hash` `06b5a7ab…77649d4`, `implementation_version` `0.1.1-reference`, `git_commit` `279bcb26…` (clean); server stopped; 0 ProofRecords; no `/verify` call; no study fixture used
Runnable	YES
Reference implementation	YES
Production implementation	NO

Open item blocking item 7 — evidence-reference status mechanism. The study policy contains no `evidence_records`, and neither the 15-field contract nor the policy defines how the verifier learns the status of an evidence reference. With the study policy loaded as-is, evidence-required requests resolve HOLD (fail closed); this is stated from the verifier code and was not tested with study fixtures. Derek Hone decision required: (a) separate hashed evidence registry via a new env var (e.g. `EP_EVIDENCE_REGISTRY`) — new verifier commit/tag, policy SHA unchanged; (b) move `evidence_records` into the study policy — policy SHA changes, packet amendment; (c) another mechanism. Nothing has been implemented. See the verifier's `POLICY_CONFIG.md` §4.

Required disclosure (verbatim):

> "The ExecutionProof implementation used in EP-AGENT-001 is a controlled reference implementation built and frozen before preregistration filing and before experimental execution. EP-AGENT-001 therefore evaluates cross-framework integration portability against this pinned reference implementation, not against an existing production deployment."